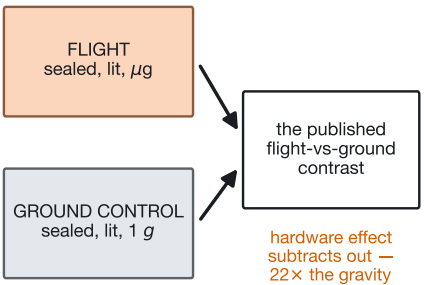
Designing the experiment that could tell hardware from gravity

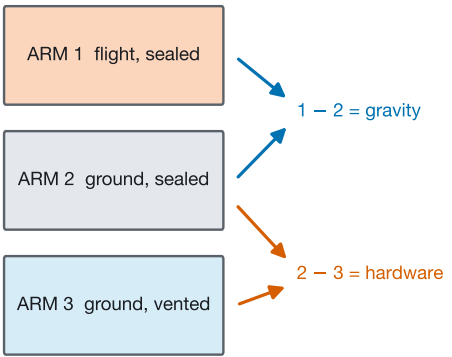
a What the standard contrast cannot see

CO_2 drawn down in BOTH arms



What survives is a persistent ~7–9 % rosette assimilation penalty at saturating light (under 2.4 % at OSD-522's), and downstream of it a carbon-starvation signature — which is what the omics show.

b The design that separates them



Three arms estimate main effects (gravity & hardware). Adding Arm 4 (flight, vented) provides the full 2×2 factorial test for a gravity×enclosure interaction.

c The gap this ran into

0 of 12

OSDR metabolomics studies that are of a plant

12 of 567

OSDR studies that are metabolite profiling, all mouse, human, rat or microbial

0 known

in-canister CO_2 logs, any BRIC-class flight

A ~20-compound targeted panel on archived material would be the first plant spaceflight metabolome.